

Task 1: Technical Exploration & Applied Research Report Template

Instructions: Complete all sections below. Keep your final submitted report strictly between 1 to 4 pages.

Intern Full Name:	Varanasi Bhavitha	Domain Track:	Full-Stack
Academic Institution:	Aditya institute of technology and management	Date of Submission:	23/08/2026
Target System Audited:	Swiggy Website	Intern ID:	VYNT-08/26-CFC02721

1. Executive Summary & Problem Scope

Describe the platform, website, or architectural workflow you chose to audit. State the primary objective of your investigation (3–4 concise sentences).

I selected the **Swiggy website** to understand how a client communicates with a server when a web page is loaded. The investigation focused on the network request, server response, HTTP status, loading time, and browser-side processing. Chrome DevTools was used to observe the communication between the browser and Swiggy servers.

2. Diagnostic Tooling & Environment Setup

List the browser version, network throttling profile, viewport resolution, and DevTools tabs utilized (e.g., Elements, Network, Console, Application).

Browser: Google Chrome (Version 151.0.0.0)
Mode: Incognito
Device: iPhone 12 Pro
Viewport: 390 × 844
Network: No throttling
DevTools: Network, Elements, Console

I used network tab to inspect request URLs, HTTP methods, status codes, response headers and timing information.

Console tab was checked for browser warnings and messages.

3. Live Audit Observations & Diagnostic Evidence

Document your empirical findings. Include API endpoint paths, HTTP status codes, latency timings (TTFB), console error logs, or paste/embed your annotated screenshots below.

I inspected the Swiggy website using Chrome DevTools to observe the communication between the browser and the Swiggy servers. In the Network tab, I observed a GET request to the `/mapi/misc_new/skeleton` API endpoint, which returned a **200 OK** status. The request took approximately **245.69 ms**, with the server response waiting time (TTFB) around **178.92 ms**. I also checked the Console, where only font resource preload warnings were observed.

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4. Deep-Dive Root Cause Analysis & Real-World Analogy

Explain the underlying technical mechanism causing the behavior/bottleneck. Connect this concept with an intuitive real-world analogy (e.g., restaurant waiter model, library desk cache, etc.).

When I opened the Swiggy website, I observed that the browser sends requests to the server to get the required information. The browser needs to wait for the server response before it can process the requested data. In my observation, the server response waiting time was 178.92 ms.

I can compare this process with ordering food in a restaurant. We place the order, the waiter takes it to the kitchen, and we wait until the food is ready. In the same way, the browser sends a request, the server processes it, and the browser waits for the response.

5. Proposed Engineering / UX Remediation

Provide your actionable solution. If it's a visual bug, show the corrected CSS rule. If it's a backend/network issue, propose schema indexing, caching, payload optimization, or input validation.

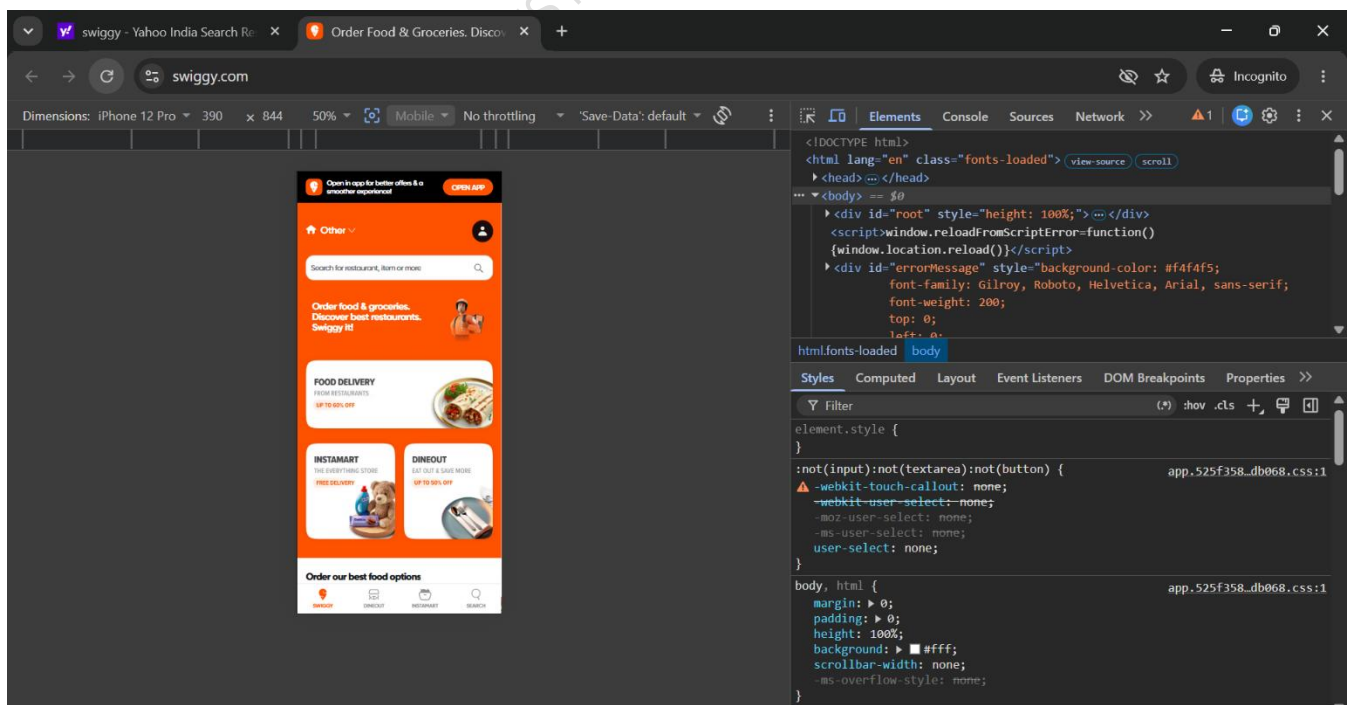
I didn't find a major problem, but I suggested some ways to improve performance. We can reduce unnecessary requests, use caching, send less data in the API response, and optimize server processing. This can reduce waiting time and make the website faster.

6. Key Learnings & 13-Week Internship Learning Roadmap

Summarize your core takeaways from this exercise and list 2-3 enterprise technologies/frameworks you aim to master during your internship at Vyntyra.

From this task, I learned how a browser communicates with a server and how requests and responses work. I also learned how to use Chrome DevTools to check network requests, status codes and response timings. This task helped me understand the basic flow of a website from the client side to the server and back.

I plan to improve my skills step by step. I will start with JavaScript and web fundamentals, then learn React.js for frontend development. After that, I will learn Node.js and Express.js for backend development and use these skills to build a small full-stack project.



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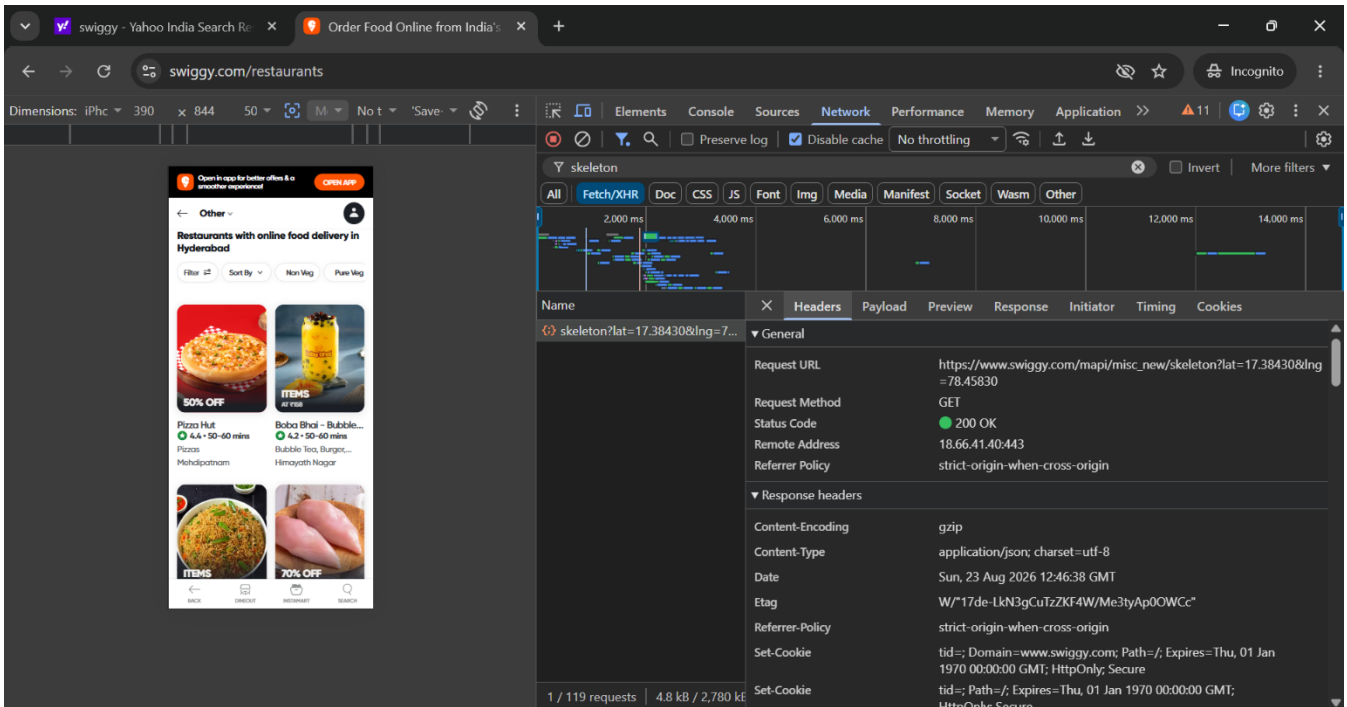


Figure 1: Network request showing GET method and 200 OK status

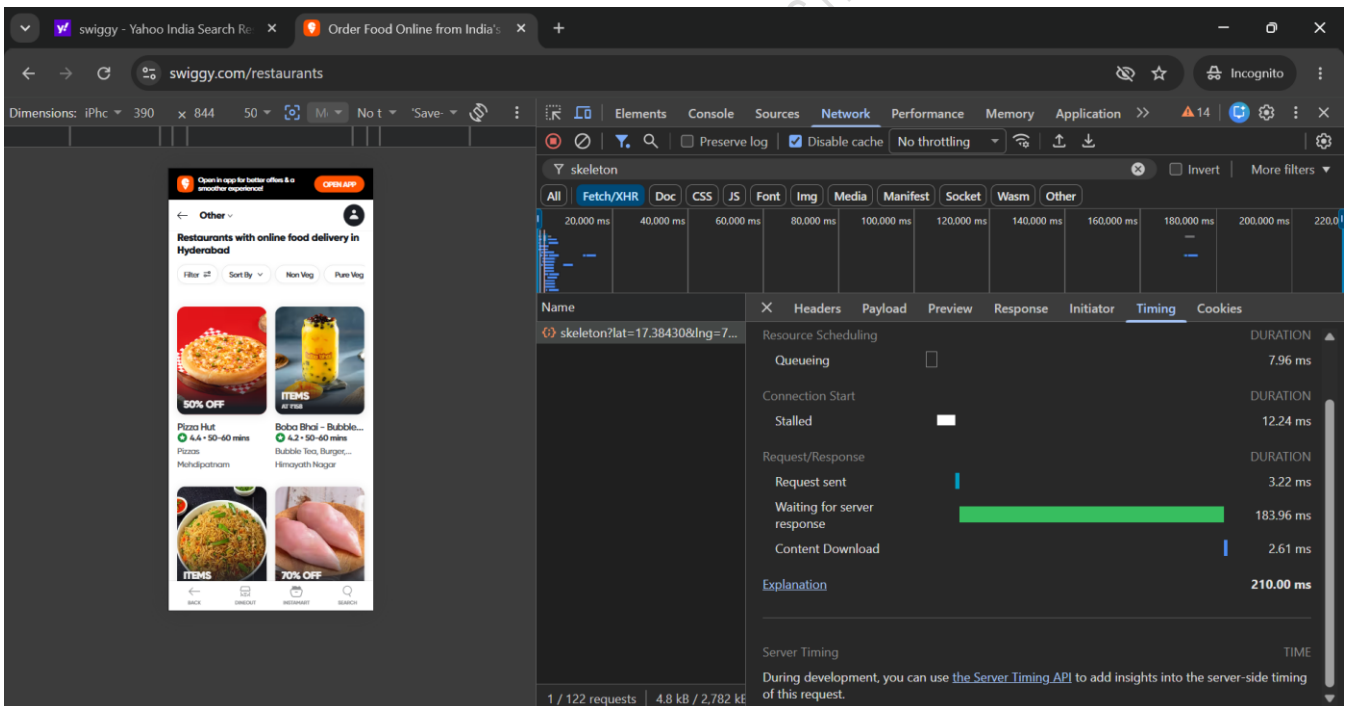


Figure 2: Network timing showing server response waiting time

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